

William J Vianese

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EDUCATION

WASHINGTON UNIVERSITY IN ST. LOUIS | St. Louis, MO

May 2026

Bachelor of Arts

Majors in Math & Computer Science and Environmental Analysis

- GPA: 3.82/4.00; Dean's List (Spring 2023, Fall 2023, Fall 2024)

Relevant Coursework

- Data Science, Data Mining, Artificial Intelligence, Analysis of Algorithms, Algorithms for Nonlinear Optimization, Data Structures, Dynamics & Computation, Financial Mathematics, Probability & Statistics, Matrix Algebra, and Applied Mathematics

TECHNICAL SKILLS

- Languages:* Python, Java, R
- Libraries & Frameworks:* Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
- Tools:* Google Suite, Microsoft Suite, GitHub, Zoom, Slack

PROJECTS

Dimensionality Reduction and Classification Analysis

- Applied principal component analysis (PCA) to reduce feature dimensionality and analyze information retention
- Implemented and compared multiple classification models such as logistic regression, k-NN, and decision trees
- Evaluated model performance using accuracy metrics and confusion matrices
- Analyzed tradeoffs between dimensionality reduction, model complexity, and predictive performance

Predictive Modeling Workflow and Model Comparison

- Built an end-to-end predictive modeling workflow including data preprocessing, feature selection, training, and evaluation
- Compared multiple predictive models to assess generalization behavior and performance
- Used quantitative metrics to support model selection and comparison
- Communicated findings through a structured technical report with supporting visualizations

Autonomous Agent Design Using Search and Reinforcement Learning

- Implemented classical search algorithms including DFS, BFS, Uniform Cost Search, and A*
- Built adversarial agents using Minimax and Alpha-Beta pruning
- Developed reinforcement learning agents via value iteration and Q-learning
- Evaluated agent behavior across environments with varying constraints and reward structures

EXPERIENCE

CAMP EMERSON | Hinsdale, MA

May 2022 – August 2025

Roles Held: Data & Systems Manager, Program Director, Wilderness Director, and Counselor

- Progressed across four roles to own and optimize core operational systems supporting 285+ users and 80+ staff, spanning enrollment, scheduling, transportation, billing, and compliance
- Designed and implemented automated workflows and process improvements to improve data consistency, reduce manual operations, and improve system reliability
- Performed data analysis and reporting on enrollment trends, logistical constraints, financial activity, and staffing needs to support leadership decisions

ACTIVITIES

WASHINGTON UNIVERSITY ROWING TEAM | St. Louis, MO

August 2022 - PRESENT

Coach and Coxswain

- Lead workouts, dieting, and recovery in conjunction with motivating and organizing the rowers
- Employ critical and quick-thinking skills to practice effectively on the water and land
- Leverage strong interpersonal skills to promote team cohesion and expand motivation

AWARDS AND RECOGNITIONS

- Selected as ACRA Third Team All American Coxswain 2024
- Selected as ACRA All Regional Coxswain in the Plains Region 2023, 2024